

Overview of Stacking ensemble

Subject: Stacking ensemble

1.

Change detection Change detection is an image analysis problem, consisting of the identification of places where the land cover has changed over time. Change detection is widely used in fields such as urban growth, forest and vegetation dynamics, land use and disaster monitoring. The earliest applications of ensemble classifiers in change detection are designed with the majority voting, Bayesian model averaging, and the maximum posterior probability. Given the growth of satellite data over time, the past decade sees more use of time series methods for continuous change detection from image stacks. One example is a Bayesian ensemble changepoint detection method called BEAST, with the software available as a package Rbeast in R, Python, and Matlab.

2.

Face recognition, which recently has become one of the most popular research areas of pattern recognition, copes with identification or verification of a person by their digital images. Hierarchical ensembles based on Gabor Fisher classifier and independent component analysis preprocessing techniques are some of the earliest ensembles employed in this field.

3.

While speech recognition is mainly based on deep learning because most of the industry players in this field like Google, Microsoft and IBM reveal that the core technology of their speech recognition is based on this approach, speech-based emotion recognition can also have a satisfactory performance with ensemble learning. It is also being successfully used in facial emotion recognition.

4.

In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. Unlike a statistical ensemble in statistical mechanics, which is usually infinite, a machine learning ensemble consists of only a concrete finite set of alternative models, but typically allows for much more flexible structure to exist among those alternatives.

5.

For each model m in the bucket: Do c times: (where ' c ' is some constant) Randomly divide the training dataset into two sets: A and B Train m with A Test m with B Select the model that obtains the highest average score

6.

Boosting involves training successive models by emphasizing training data mis-classified by previously learned models. Initially, all data (D_1) has equal weight and is used to learn a base model M_1 . The examples mis-classified by M_1 are assigned a weight greater than correctly classified examples. This boosted data (D_2) is used to train a second base model M_2 , and so on. Inference is done by voting. In some cases, boosting has yielded better accuracy than bagging, but tends to over-fit more.

The most common implementation of boosting is Adaboost, but some newer algorithms are reported to achieve better results.

7.

Intrusion detection An intrusion detection system monitors computer network or computer systems to identify intruder codes like an anomaly detection process. Ensemble learning successfully aids such monitoring systems to reduce their total error.

8.

Amended Cross-Entropy Cost: An Approach for Encouraging Diversity in Classification Ensemble
The most common approach for training classifier is using Cross-entropy cost function. However, one would like to train an ensemble of models that have diversity so when we combine them it would provide best results. Assuming we use a simple ensemble of averaging K classifiers. Then the Amended Cross-Entropy Cost is

9.

Land cover mapping Land cover mapping is one of the major applications of Earth observation satellite sensors, using remote sensing and geospatial data, to identify the materials and objects which are located on the surface of target areas. Generally, the classes of target materials include roads, buildings, rivers, lakes, and vegetation. Some different ensemble learning approaches based on artificial neural networks, kernel principal component analysis (KPCA), decision trees with boosting, random forest and automatic design of multiple classifier systems, are proposed to efficiently identify land cover objects.

10.

Ensemble size While the number of component classifiers of an ensemble has a great impact on the accuracy of prediction, there is a limited number of studies addressing this problem. A priori determining of ensemble size and the volume and velocity of big data streams make this even more crucial for online ensemble classifiers. Mostly statistical tests were used for determining the proper number of components. More recently, a theoretical framework suggested that there is an ideal number of component classifiers for an ensemble such that having more or less than this number of classifiers would deteriorate the accuracy. It is called "the law of diminishing returns in ensemble construction." Their theoretical framework shows that using the same number of independent component classifiers as class labels gives the highest accuracy.

11.

Stacking Stacking (sometimes called stacked generalization) involves training a model to combine the predictions of several other learning algorithms. First, all of the other algorithms are trained using the available data, then a combiner algorithm (final estimator) is trained to make a final prediction using all the predictions of the other algorithms (base estimators) as additional inputs or using cross-validated predictions from the base estimators which can prevent overfitting. If an arbitrary combiner algorithm is used, then stacking can theoretically represent any of the ensemble techniques described in this article, although, in practice, a logistic regression model is often used as the combiner. Stacking typically yields performance better than any single one of the trained models. It has been successfully

used on both supervised learning tasks (regression, classification and distance learning) and unsupervised learning (density estimation). It has also been used to estimate bagging's error rate. It has been reported to out-perform Bayesian model-averaging. The two top-performers in the Netflix competition utilized blending, which may be considered a form of stacking. Bayesian predictive stacking generalizes the idea of stacking from the statistical estimation literature to the combination of posterior predictive distributions. These ideas have also been developed and investigated for Gaussian process models, especially for spatial data analysis and can be used to construct transfer learning frameworks for massive spatial data sets.

12.

Overview Supervised learning algorithms search through a hypothesis space to find a suitable hypothesis that will make good predictions with a particular problem. Even if this space contains hypotheses that are very well-suited for a particular problem, it may be very difficult to find a good one. Ensembles combine multiple hypotheses to form one which should be theoretically better. Ensemble learning trains two or more machine learning algorithms on a specific classification or regression task. The algorithms within the ensemble model are generally referred as "base models", "base learners", or "weak learners" in literature. These base models can be constructed using a single modelling algorithm, or several different algorithms. The idea is to train a diverse set of weak models on the same modelling task, such that the outputs of each weak learner have poor predictive ability (i.e., high bias), and among all weak learners, the outcome and error values exhibit high variance. Fundamentally, an ensemble learning model trains at least two high-bias (weak) and high-variance (diverse) models to be combined into a better-performing model. The set of weak models — which would not produce satisfactory predictive results individually — are combined or averaged to produce a single, high performing, accurate, and low-variance model to fit the task as required. Ensemble learning typically refers to bagging (bootstrap aggregating), boosting or stacking/blending techniques to induce high variance among the base models. Bagging creates diversity by generating random samples from the training observations and fitting the same model to each different sample — also known as homogeneous parallel ensembles. Boosting follows an iterative process by sequentially training each base model on the up-weighted errors of the previous base model, producing an additive model to reduce the final model errors — also known as sequential ensemble learning. Stacking or blending consists of different base models, each trained independently (i.e. diverse/high variance) to be combined into the ensemble model — producing a heterogeneous parallel ensemble. Common applications of ensemble learning include random forests (an extension of bagging), Boosted Tree models, and Gradient Boosted Tree Models. Models in applications of stacking are generally more task-specific — such as combining clustering techniques with other parametric and/or non-parametric techniques. Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model. In one sense, ensemble learning may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation. On the other hand, the alternative is to do a lot more learning with one non-ensemble model. An ensemble may be more efficient at improving overall accuracy for the same increase in compute, storage, or communication resources by using that increase on two or more methods, than would have been improved by increasing resource use for a single method. Fast algorithms such as decision trees are commonly used in ensemble methods (e.g., random forests), although slower algorithms can benefit from ensemble techniques as well. By analogy, ensemble techniques have been used also in unsupervised learning scenarios, for example in consensus clustering or in anomaly detection.